

A reusable RAG pipeline

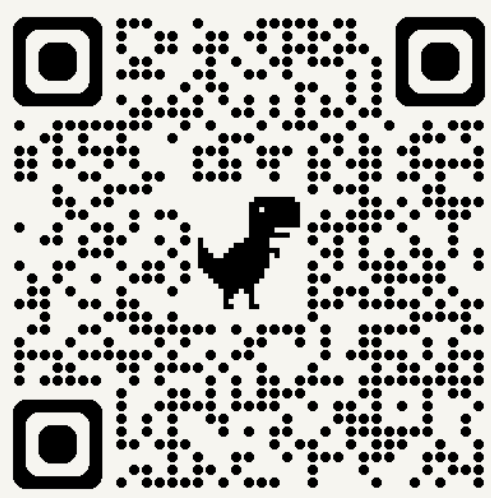
A BioASQ reproduction kit

A model-organism case study

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available digitally.
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Top-ranked retrieval in BioASQ 14b Phase A

Ranked 1st in document retrieval in 3 of 4 batches · preliminary results

01 A reusable RAG pipeline

github.com/fulaibaowang/RAG-scripts

Configurable

Pipeline stages and settings
in one config

Reproducible

Docker image,
pinned dependencies

Measurable

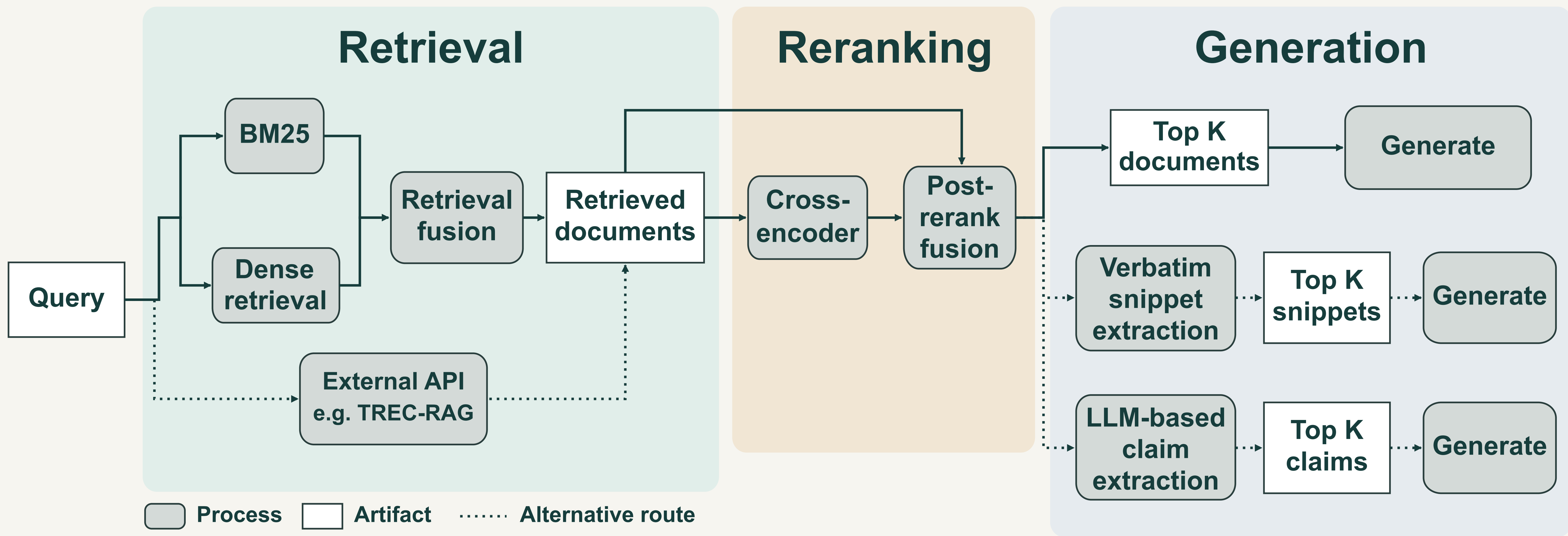
Metrics and plots
at every stage

Portable

BioASQ, TREC-RAG,
your corpus

Agent-friendly

Built-in guidance for
AI coding assistants



02 A BioASQ reproduction kit

github.com/fulaibaowang/BioASQ

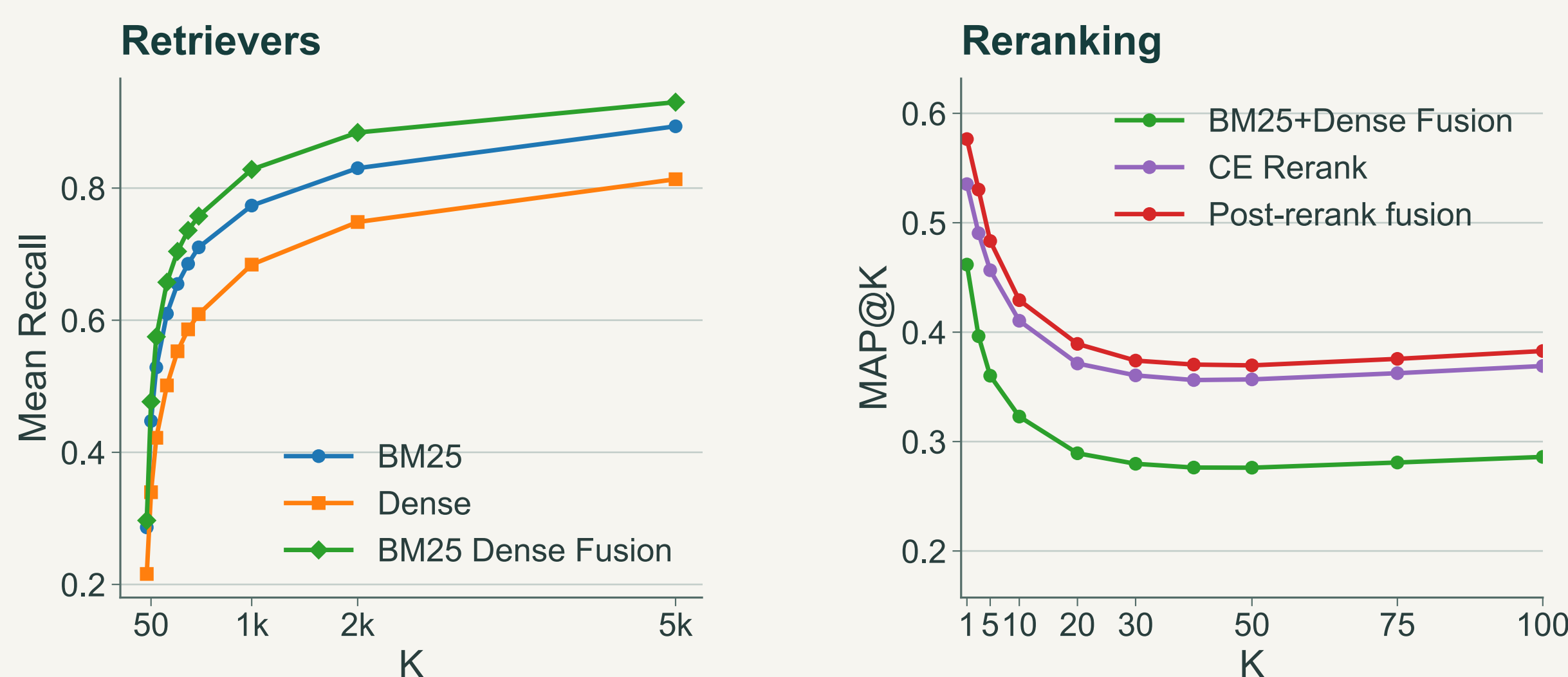
Scripts and configurations to reproduce our BioASQ retrieval runs.



One batch of 80 questions · one A100 GPU

16 CPU cores **~80 GB** RAM **~1.5 h** retrieve + rerank **~150 GB** one-time: corpus, indexes, container

Automatic diagnostic plots · detailed analysis in the working note



Examples on BioASQ 13b, batches 1–4 averaged

BEYOND BIOASQ

03 A model-organism case study

github.com/fulaibaowang/dictycite

Why is our system called DictyCite?

Dicty (*Dictyostelium*) is a model organism in biology. In a separate study, we used this pipeline to benchmark literature retrieval for it.

Our *Dictyostelium* case study suggests that tailoring literature retrieval to a biological domain can be worthwhile, using curated annotations to enrich queries and choosing retrievers and rerankers that fit the task.

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Benchmarking Literature Retrieval for a Model Organism: A Dictyostelium Case Study

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